

Solving the Phase Unwrapping Problem in Dual Divergent Projection Systems Using a Physics-Informed Neural Network

Jorge R. Parra-Michel
Facultad de Ingeniería
Universidad De La Salle Bajío
León, Guanajuato, Mexico
jrparra@delasalle.edu.mx

Nelson A. Martínez Santamaría
Facultad de Ingeniería
Universidad De La Salle Bajío
León, Guanajuato, Mexico
nms107283@lasallebajio.edu.mx

Enrique Aguilar Vargas
Facultad de Ingenierías y Tecnologías
Universidad De La Salle Bajío
León, Guanajuato, Mexico
eaguilar@lasallebajio.edu.mx

Abstract—Three-dimensional surface reconstruction based on dual divergent structured-light projection produces a spatially variant, nonlinear phase-to-height relationship that makes conventional phase unwrapping algorithms unreliable. This paper presents the application of a Physics-Informed Hybrid U-Net Neural Network (PIHUNN) to solve the phase unwrapping problem directly from wrapped phase maps acquired in such systems. The proposed approach bypasses explicit phase unwrapping by learning a direct mapping from the observed wrapped phase to the surface height, embedding physical constraints—specifically, a gradient bound derived from the optical model $\|\nabla\hat{h}\| \leq 2\pi/\lambda_{\min}$ —into a hybrid loss function. The network architecture incorporates a frequency decomposition front-end, pre-activated residual blocks, and squeeze-excitation channel attention to handle the multifrequency content inherent in moiré fringe patterns. Experimental results on both synthetic and real data demonstrate that PIHUNN achieves an RMSE of 0.053 ± 0.024 (normalized units), $R^2 = 0.974$, and SSIM of 0.942, outperforming standard U-Net, Pix2Pix GAN, Zernike fitting, and branch-cut methods while maintaining real-time inference at 12.3 ms per frame on an NVIDIA T4 GPU.

Keywords—phase unwrapping, structured light, deep learning, physics-informed neural network, 3D reconstruction, moiré fringe analysis

I. Introduction

Structured-light projection is a well-established technique for noncontact three-dimensional (3D) surface measurement with applications spanning industrial inspection, reverse engineering, and biomedical imaging [1], [2]. In its conventional single-projector configuration, the relationship between the observed phase and the object height is typically linear or mildly nonlinear, and established phase unwrapping algorithms—spatial, temporal, or hybrid—provide reliable results [3], [4].

This work was supported by SECIHTI, Mexico, through the Sistema Nacional de Investigadores program, the Santander-FIMPES Research Grant Program 2025 and La Salle Bajío University. An extended version of this work with complete methodology and ablation studies is available in [11].

However, when the measurement employs a dual divergent projection setup, the phase-to-height mapping becomes fundamentally more challenging. In this configuration, two projectors illuminate the object from different angles using divergent fringe patterns, generating a moiré-like interference field on the surface. The resulting phase sensitivity is spatially variant and nonlinear, and conventional unwrapping strategies often fail in regions with steep gradients, shadows, or discontinuities [5]–[7].

Recent advances in deep learning for optical metrology have shown that convolutional neural networks can learn robust mappings for phase demodulation and unwrapping under difficult conditions [8], [9]. Nevertheless, most existing approaches are designed for single-projector or collimated illumination setups and do not explicitly incorporate the spatially varying physics that governs dual divergent projection.

This paper demonstrates the application of a Physics-Informed Hybrid U-Net Neural Network (PIHUNN) [11] to solve the phase unwrapping problem in dual divergent projection systems. The key contributions are: (1) a detailed description of how the PIHUNN architecture and its physics-informed training procedure are employed to handle the nonlinear inverse problem posed by dual divergent illumination; (2) a comprehensive analysis of the processing pipeline from wrapped phase acquisition to 3D height recovery; and (3) quantitative benchmarking against classical and learning-based methods on both synthetic and experimental data.

The remainder of this paper is organized as follows. Section II formulates the phase-to-height problem under dual divergent projection. Section III describes the proposed PIHUNN architecture and physics-informed loss function. Section IV details the training protocol. Section V presents quantitative and qualitative results. Section VI concludes the paper.

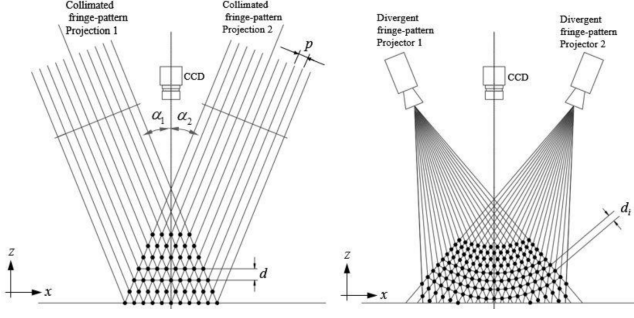


Fig. 1. Double-projector fringe-projection geometry. Two fringe-pattern projectors illuminate the surface from distinct directions (collimated or divergent configurations), while a single CCD camera records the resulting fringe deformation. The projection angles (e.g., α_1 and α_2) and the fringe pitch (e.g., p) determine the phase-to-height sensitivity, whereas the spatial separation and viewing geometry influence occlusions, shadow regions, and overall reconstruction stability. This configuration provides complementary illumination that mitigates missing information in regions where a single projection would be shadowed or poorly observed.

II. Problem Formulation

Throughout this paper, the in-plane pixel coordinate is denoted by $\mathbf{x} = (x, y)$, the surface height by $h(\mathbf{x})$, the metric phase by $\varphi_m(\mathbf{x})$, and the wrapped phase by $\varphi_w(\mathbf{x}) \in (-\pi, \pi]$.

A. Dual Divergent Projection Setup

The optical configuration consists of two structured-light projectors P_1 and P_2 positioned symmetrically about the optical axis of a CCD camera at positions $S_1(-s, 0, l)$ and $S_2(s, 0, l)$. Each projector emits a divergent sinusoidal fringe pattern, and the camera records the intensity field reflected by the surface. The dual-projection geometry improves robustness against shadowing but introduces a spatially varying sensitivity because the effective fringe period changes with range and direction.

Figure 1 illustrates the measurement geometry of the double-projection fringe setup considered in this work. Two projectors illuminate the object from different directions while a single camera observes the deformed fringe patterns, which enables complementary viewing angles and improves robustness against shadowing and discontinuities. The diagram defines the key geometric quantities used later in the forward model, including the projection angles and the effective fringe pitch, which jointly determine the sensitivity of phase to height variations.

B. Phase-to-Height Relationship

In moiré fringe projection, the observed phase is derived from the relative deformation of projected fringe patterns. For collimated illumination, the moiré fringe spacing is constant and the phase-to-height relationship is linear:

$$\Delta\varphi = \frac{4\pi}{p_L} \sin\left(\frac{\alpha_L}{2}\right) \Delta z, \quad (1)$$

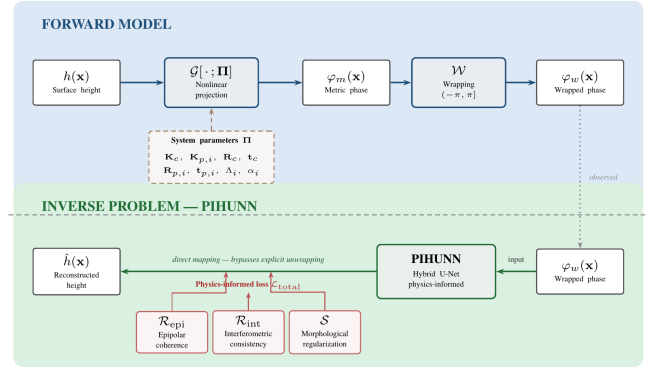


Fig. 2. Problem formulation. (Top) Forward model: the surface height $h(\mathbf{x})$ is mapped to metric phase $\varphi_m(\mathbf{x})$ through a nonlinear projection $\mathcal{G}[\cdot; \Pi]$ defined by the system parameters Π , and then wrapped by $\mathcal{W}(\cdot)$ to yield the observed wrapped phase $\varphi_w(\mathbf{x}) \in (-\pi, \pi]$. (Bottom) Inverse problem solved by PIHUNN: a direct mapping $\varphi_w(\mathbf{x}) \mapsto \hat{h}(\mathbf{x})$ is learned, bypassing explicit phase unwrapping. Training is regularized using a physics-informed loss $\mathcal{L}_{\text{total}}$ combining gradient fidelity normalized by the optical bound $\|\nabla h\| \leq 2\pi/\lambda_{\min}$, multi-scale gradient consistency, and total variation regularization.

where p_L is the fringe period and α_L is the angle between the illuminations. Under divergent illumination, however, the fringe period varies with position, producing a spatially variant and nonlinear sensitivity.

C. Forward Model and Inverse Problem

The measurement can be described by a forward model:

$$\varphi_w(\mathbf{x}) = \mathcal{W}(\varphi_m(\mathbf{x})), \quad \varphi_m(\mathbf{x}) = \mathcal{G}(h(\mathbf{x}); \Pi), \quad (2)$$

where $\mathcal{G}[\cdot; \Pi]$ denotes the nonlinear projection governed by system parameters Π , and $\mathcal{W}(\cdot)$ is the wrapping operator mapping phase to $(-\pi, \pi]$. In the proposed approach, the inverse problem is solved by directly learning a mapping f_θ such that:

$$\hat{h}(\mathbf{x}) = f_\theta(\varphi_w(\mathbf{x})), \quad (3)$$

bypassing explicit phase unwrapping while enforcing physics-based regularization during training.

Figure 2 formalizes the measurement process and the inverse task addressed in this work. The forward pipeline maps surface height to metric phase through the system-dependent nonlinear projection and then applies wrapping to obtain the observed wrapped phase. The lower panel motivates the proposed learning strategy: PIHUNN directly estimates height from the wrapped observation, thereby avoiding explicit phase unwrapping while enforcing physical consistency through structured regularization terms. This formulation provides the conceptual bridge between interferometric physics and data-driven inference.

III. Proposed Method

A. Network Architecture

PIHUNN is built upon a symmetric U-Net encoder-decoder with four encoder stages (32–64–128–256 filters)

plus a 512-channel bottleneck and three architectural innovations tailored to moiré fringe processing:

Frequency Decomposition Front-end (FDDS). The input wrapped phase map $\varphi_w(\mathbf{x})$ is processed by three parallel convolutional branches with kernels 7×7 , 3×3 , and 1×1 , each followed by batch normalization and ReLU. The resulting feature maps are concatenated:

$$f_{\text{low}} = \text{ReLU}(\text{BN}(\text{Conv}_{7 \times 7}(\varphi_w))), \quad (4)$$

$$f_{\text{high}} = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\varphi_w))), \quad (5)$$

$$f_{\text{id}} = \text{ReLU}(\text{BN}(\text{Conv}_{1 \times 1}(\varphi_w))), \quad (6)$$

$$f_{\text{FDDS}} = \text{Concat}(f_{\text{low}}, f_{\text{high}}, f_{\text{id}}). \quad (7)$$

Residual Pre-activation Blocks. Each U-Net level uses pre-activation residual blocks to improve gradient flow and stabilize training for deep feature hierarchies.

Squeeze-and-Excitation (SE) Attention. SE modules are inserted at deeper encoder levels to recalibrate channel-wise feature responses. The global average pooling produces $z_c = \frac{1}{HW} \sum_{i,j} u_c(i,j)$, followed by a two-layer MLP with reduction ratio $r=16$ that computes attention weights $\mathbf{s} = \sigma_{\text{sig}}(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z}))$, which rescale each channel as $\tilde{u}_c = s_c \cdot u_c$.

The decoder recovers spatial resolution via transposed convolutions, concatenates skip connections from the corresponding encoder level, and applies progressive dropout (0.2–0.4) to control effective capacity near the output. The final layer applies a sigmoid activation to bound $\hat{h} \in [0, 1]$, consistent with the min–max normalization applied to the training targets:

$$\tilde{h}(\mathbf{x}) = \frac{h(\mathbf{x}) - h_{\min}}{h_{\max} - h_{\min}}, \quad \hat{h}(\mathbf{x}) \approx \tilde{h}(\mathbf{x}) \in [0, 1]. \quad (8)$$

Here, $h_{\min}$ and $h_{\max}$ define the calibrated height range. During inference, the physical estimate is recovered as $\hat{h}_{\text{phys}}(\mathbf{x}) = h_{\min} + (h_{\max} - h_{\min}) \hat{h}(\mathbf{x})$.

Figure 3 introduces the proposed PIHUNN architecture, which is designed to estimate the surface height map directly from a wrapped-phase input by combining multi-scale feature extraction with a U-Net-style reconstruction pipeline. Since wrapped phase inherently contains 2π discontinuities and localized high-frequency variations, a parallel-branch front-end is employed to capture complementary spatial structures before a symmetric encoder-decoder aggregates global context while preserving fine details through skip connections. The decoder then produces a normalized height estimate that is finally denormalized to recover physical units, providing a consistent inference route suitable for high-gradient optical metrology scenarios.

B. Physics-Informed Loss Function

The training objective combines four terms that enforce physical consistency:

$$\mathcal{L}_{\text{total}} = \omega_1 \mathcal{L}_{\text{MSE}} + \omega_2 (1 - \text{SSIM}) + \omega_3 \mathcal{L}_{\nabla} + \omega_4 \mathcal{L}_{\text{perc}}, \quad (9)$$

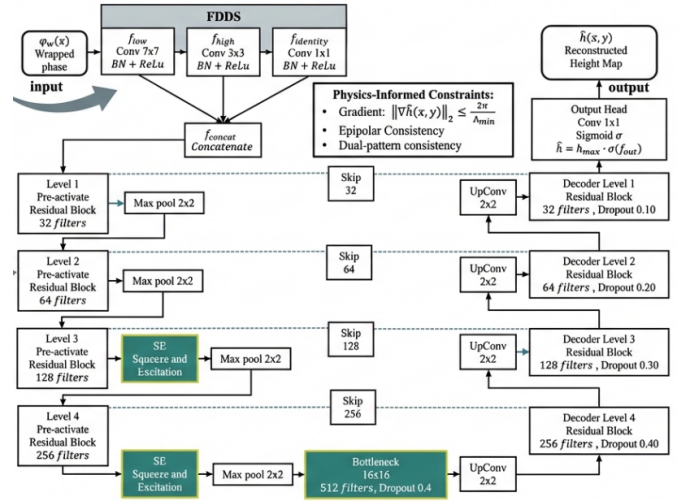


Fig. 3. Architecture of the Physics-Informed Hybrid U-Net Neural Network (PIHUNN) for direct height estimation from wrapped phase. The input is the normalized wrapped-phase map $\varphi_w(\mathbf{x}) \in [0, 1]$, processed by a frequency-decomposition front-end using three parallel convolutional branches (kernels 7×7 , 3×3 , and 1×1) whose feature maps are concatenated to capture complementary spatial scales. A U-Net backbone with pre-activation residual blocks performs multi-resolution encoding (32–64–128–256 filters) and decoding with skip connections. Squeeze-and-Excitation (SE) modules are introduced at deeper levels to recalibrate channel responses, while progressive dropout in the decoder improves generalization. The output head produces the reconstructed height map $\hat{h}(\mathbf{x})$ via a 1×1 convolution followed by a sigmoid and final scaling to the calibrated physical range. Note: physics-informed constraints are applied during training via the loss function, not as architectural modules.

where the weights $(\omega_1, \omega_2, \omega_3, \omega_4) = (1.0, 0.2, 0.05, 0.02)$ are chosen to balance the expected magnitude of each term on the validation set (so that no single component dominates the gradient), and kept fixed thereafter. The perceptual feature extractor $\Phi_\ell(\cdot)$ is implemented as a fixed, pretrained convolutional network (VGG-16), evaluated on the single-channel maps replicated to three channels.

- $\mathcal{L}_{\text{MSE}}$ ensures pointwise metric accuracy between $\hat{h}$ and h^{GT} .
- $(1 - \text{SSIM})$ preserves mesoscale structural coherence of the reconstructed surface relief.
- $\mathcal{L}_{\nabla}$ constrains gradient magnitude, normalized by the physics-derived bound $2\pi/\lambda_{\min}$ from the optical model, to maintain physically plausible slopes.
- $\mathcal{L}_{\text{perc}}$ improves perceptual fidelity by matching deep feature activations.

The physics-informed character of the gradient term derives from the phase formation model: $\phi_k(x, y) = (2\pi/\lambda_{\text{eff},k}) d_k(x, y)$, which implies $\|\nabla \phi_k\| \leq 2\pi/\lambda_{\min}$. This bound is used to normalize the gradient fidelity loss, ensuring predicted surfaces cannot exhibit gradients exceeding the optical limit imposed by the minimum fringe period.

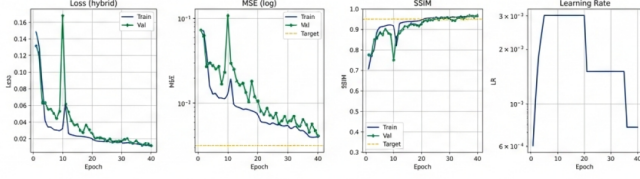


Fig. 4. Training dynamics of PIHUNN. The hybrid loss decreases consistently for both training and validation, indicating stable optimization. The logarithmic MSE shows convergence toward the target range, while SSIM increases and saturates, reflecting improved structural fidelity of the reconstructed height maps. The learning-rate schedule (right) shows the staged reduction used to refine convergence after the initial rapid descent. Training performed on NVIDIA T4/A100 GPUs.

TABLE I
Comparative Performance on Test Set (Normalized Units)

Method	RMSE (norm. [0, 1])	R^2	SSIM	Time (ms)
PIHUNN	0.053 ± 0.024	0.974	0.942	12.3 [†]
U-Net (standard)	0.078 ± 0.031	0.953	0.908	11.8
Pix2Pix GAN	0.092 ± 0.038	0.931	0.885	28.6
Zernike fitting	0.147 ± 0.052	0.876	0.792	156.2
Branch-cut	0.189 ± 0.067	0.812	0.743	234.5

IV. Training and Implementation

The training pipeline uses paired synthetic and experimental samples. Synthetic data are generated using a digital-twin forward model that simulates dual divergent fringe projection, followed by standard demodulation to obtain wrapped phase maps. The dataset comprises 4,000 phase–height pairs at 256×256 resolution, split 70%/15%/15% for training/validation/testing. Experimental samples are acquired using the physical setup and processed identically to produce $\varphi_w(\mathbf{x})$ inputs. Training uses Adam optimization with an initial learning rate of 3×10^{-3} , batch size 64, and early stopping based on validation SSIM. Training dynamics are shown in Fig. 4.

V. Results and Discussion

A. Quantitative Evaluation

Table I summarizes the comparative performance of PIHUNN against both classical and learning-based methods on the test set. All metrics are computed on the normalized height range [0, 1]; for a typical measurement volume of 50 mm depth, the reported RMSE corresponds to approximately 2.7 mm absolute error.

PIHUNN achieves the lowest RMSE (0.053 ± 0.024) with $R^2 = 0.974$ and $SSIM = 0.942$, while maintaining real-time inference at 12.3 ms. The standard U-Net, which shares a similar encoder-decoder topology but lacks the frequency decomposition front-end, SE attention, and physics-informed loss, performs competitively but exhibits a consistent accuracy gap. Classical approaches (Zernike

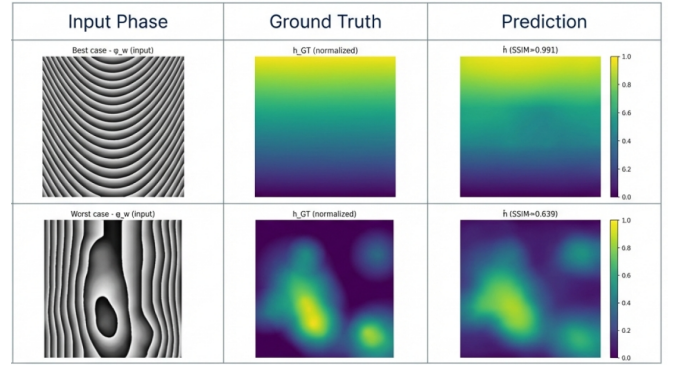


Fig. 5. Qualitative reconstruction results for representative cases. Left: input wrapped phase φ_w (normalized). Middle: ground-truth height h_{GT} (normalized). Right: PIHUNN prediction $\hat{h}$ (normalized), annotated with SSIM. The best-case example shows near-perfect structural recovery and smooth gradients consistent with the reference, whereas the worst-case example highlights the degradation under strong phase discontinuities and complex morphology, where residual smoothing and localized errors remain visible.

fitting and branch-cut unwrapping) show reduced robustness under dual divergent sensitivity, leading to higher error and longer execution time.

Note that temporal (multi-frequency) phase unwrapping methods were not included because they require sequential projections at multiple frequencies, which is incompatible with the single-shot dual-projection configuration addressed in this work. Similarly, PhaseNet [10] solves the distinct problem of phase unwrapping (wrapped $\rightarrow$ unwrapped phase), whereas PIHUNN addresses the phase-to-height inverse problem directly.

B. Qualitative Results

Figure 5 presents representative qualitative reconstructions to illustrate the behavior of PIHUNN under favorable and challenging conditions. The comparison between input wrapped phase, ground-truth height, and predicted height makes the reconstruction fidelity visually explicit, while the reported SSIM values provide a compact quantitative reference. The best-case example reflects near-ideal structural recovery, whereas the worst-case example exposes the limitations that arise in the presence of severe discontinuities and complex morphology.

Figure 6 extends the qualitative assessment with a broader set of fringe patterns and their corresponding reconstructed surfaces. The examples are intended to demonstrate consistency across varying fringe densities, localized distortions, and different surface topologies, which are commonly encountered in practical interferometric measurements. This visual gallery complements the representative cases by providing evidence of robustness across diverse input conditions.

VI. Conclusion

This paper presented PIHUNN, a physics-informed hybrid U-Net designed to solve the phase unwrapping

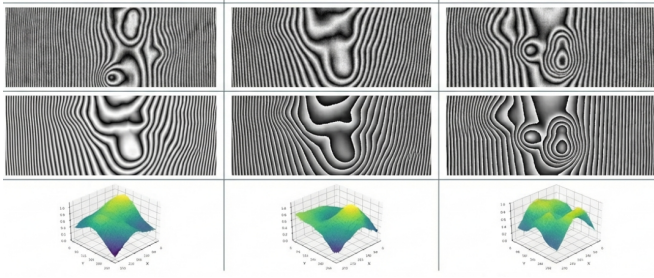


Fig. 6. Additional qualitative examples. Top and middle rows: wrapped fringe patterns corresponding to different surface morphologies and discontinuity layouts. Bottom row: reconstructed height surfaces (normalized) produced by PIHUNN. The examples illustrate the capability to recover global shape while preserving dominant features under varying fringe densities and localized distortions, which are typical in practical interferometric measurements.

problem in dual divergent structured-light projection systems by learning a direct mapping from wrapped phase to surface height. The proposed architecture combines a frequency decomposition front-end, residual encoding-decoding, and SE attention to handle multifrequency moiré content, while the physics-informed loss—specifically, the gradient fidelity term normalized by the optical bound $\|\nabla \hat{h}\| \leq 2\pi/\lambda_{\min}$ —improves consistency and generalization. Experimental evaluations on synthetic and real data show that PIHUNN outperforms classical unwrapping and baseline deep models while maintaining real-time inference. Complete methodology, ablation studies, and robustness analysis are available in the extended journal version [11].

Acknowledgments

This conference paper presents a condensed version of the methodology developed in our extended journal article [11]. The complete architecture description, ablation studies, and robustness analysis are available at <https://doi.org/10.1016/j.optlaseng.2026.109693>. This research was partially supported by the Santander-FIMPES Research Grant Program through the project “Desarrollo y Evaluación de Competencias en Manufactura Avanzada: Aplicaciones de IA en Inspección Óptica, Ingeniería Inversa y Análisis Computacional.” and the Universidad La Salle Bajío.

References

- [1] S. Van der Jeught and J. J. J. Dirckx, “Real-time structured light profilometry: a review,” *Opt. Lasers Eng.*, vol. 87, pp. 18–31, 2016.
- [2] S. Feng, C. Zuo, L. Zhang, T. Tao, Y. Hu, W. Yin, J. Qian, and Q. Chen, “Calibration of fringe projection profilometry: a comparative review,” *Opt. Lasers Eng.*, vol. 143, p. 106622, 2021.
- [3] Z. Wang, H. Du, and H. Bi, “Out-of-plane shape determination in generalized fringe projection profilometry,” *Opt. Express*, vol. 14, no. 25, pp. 12122–12133, 2006.
- [4] Z. Wang, H. Du, S. Park, and H. Xie, “Three-dimensional shape measurement with a fast and accurate approach,” *Appl. Opt.*, vol. 48, no. 6, pp. 1052–1061, 2009.
- [5] J. R. Parra-Michel, R. Martínez-Peláez, and A. Duarte-Moller, “Double structured light with divergent projection for surface topometry,” *Meas. Sci. Technol.*, vol. 32, no. 9, p. 095205, 2021.
- [6] D. Purcell and R. Cohn, “Effective wavelength calibration for moiré fringe projection,” *Appl. Opt.*, vol. 45, no. 34, pp. 8629–8635, 2006.
- [7] S. Ordonez, M. Servin, and J. S. Kang, “Moiré profilometry through simultaneous dual fringe projection for accurate phase demodulation: a comparative study,” *Appl. Opt.*, vol. 60, no. 28, pp. 8667–8675, 2021.
- [8] C. Zuo, J. Qian, S. Feng, W. Yin, Y. Li, P. Fan, J. Han, K. Qian, and Q. Chen, “Deep learning in optical metrology: a review,” *Light: Sci. Appl.*, vol. 11, p. 39, 2022.
- [9] W. Yin, Y. Che, X. Li, M. Li, Y. Hu, S. Feng, E. Y. Lam, Q. Chen, and C. Zuo, “Physics-informed deep learning for fringe pattern analysis,” *Opto-Electron. Adv.*, vol. 7, no. 1, p. 230034, 2024.
- [10] G. E. Spoorthi, S. Gorthi, and R. K. S. S. Gorthi, “PhaseNet: a deep convolutional neural network for two-dimensional phase unwrapping,” *IEEE Signal Process. Lett.*, vol. 26, no. 1, pp. 54–58, 2019.
- [11] J. R. Parra-Michel, E. Ovalle-Magallanes, and E. Aguilar-Vargas, “Physics-informed hybrid U-Net architecture for phase-to-height reconstruction in dual divergent structured-light systems,” *Opt. Lasers Eng.*, vol. 201, p. 109693, 2026. [Online]. Available: <https://doi.org/10.1016/j.optlaseng.2026.109693>
- [12] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, “Physics-informed machine learning,” *SIAM Rev.*, vol. 63, no. 3, pp. 1–76, 2021.
- [13] Z. Huang, J. Xi, Y. Yu, Q. Guo, and L. Song, “Improved geometrical model of fringe projection profilometry,” *Opt. Express*, vol. 22, no. 26, pp. 32220–32232, 2014.
- [14] H. Du and Z. Wang, “Three-dimensional shape measurement with an arbitrarily arranged fringe projection profilometry system,” *Opt. Lett.*, vol. 32, no. 16, pp. 2438–2440, 2007.
- [15] J. Lu, R. Mo, H. Sun, and Z. Chang, “Flexible calibration of phase-to-height conversion in fringe projection profilometry,” *Appl. Opt.*, vol. 55, no. 23, pp. 6381–6388, 2016.